

Context



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Source: A. Torralba

Humans work with strong priors

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Source: A. Torralba

Conclusion

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Human vision relies not **only** on data term (pixel information)

Priors are important

They explain the **most likely** scene if data is ambiguous:

$$\operatorname{argmax}_y P(y|I) = \operatorname{argmax}_y P(I|y)P(y)$$

Bayes formula:

$$P(y|I) = \frac{P(I|y)P(y)}{P(I)}$$

Posterior = Likelihood x Prior

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Image as function

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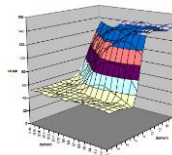
45	60	98	127	132	133	137	133
46	65	98	125	126	128	131	135
47	65	96	115	116	123	135	137
47	63	91	107	113	122	138	134
50	59	80	97	110	123	133	134
49	55	68	83	97	113	128	133
50	50	58	70	84	102	116	126
50	50	52	58	69	86	101	120

Mapping from image domain (pixels)
to values

Gray image: $f: \Omega \rightarrow \mathbb{R}$

Color images: $f: \Omega \rightarrow \mathbb{R}^3$

Image represented as matrix or
tensor



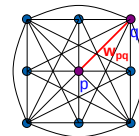
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Images as graphs

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- Fully-connected graph
 - node (vertex) for every pixel
 - link between every pair of pixels, p, q
 - affinity weight w_{pq} for each link (edge)
 - w_{pq} measures *similarity*
 - similarity is *inversely proportional* to difference (in color and position...)

Source: S. Seitz

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Histogram equalization

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Pixels as distribution:

$$p_f(y) = \frac{|\{x \in \Omega : f(x) = y\}|}{|\Omega|}$$

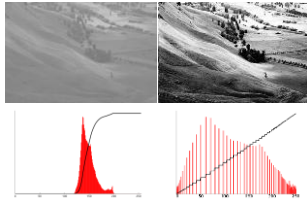
$$F_f(y) = \int_0^y p_f(y') dy' = \sum_{y'=0}^y p_f(y')$$

Cumulative distribution function

Make CDF linear:

$$h(f(x)) = F(f(x)) \cdot \max_{x \in \Omega} f(x)$$

$$p_h(z) = \frac{|\{x \in \Omega : h(f(x)) = z\}|}{|\Omega|} = \frac{1}{\max_{x \in \Omega} f(x)}$$



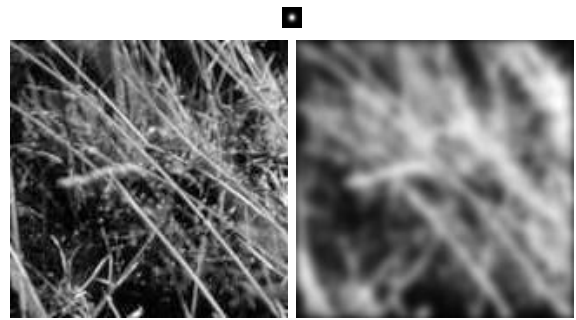
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Linear filtering

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Source: S. Lazebnik

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Defining convolution

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Let f be the image and h be the kernel. The output of convolving f with h is denoted: $g = f * h$

$$g(i, j) = \sum_{k, l} f(i - k, j - l) h(k, l) = \sum_{k, l} f(k, l) h(i - k, j - l)$$

45	60	98	127	132	133	137	133
46	65	96	123	126	128	131	133
47	65	96	115	119	123	135	137
47	63	91	107	113	122	138	134
50	59	89	97	110	123	133	134
49	53	68	83	97	113	128	133
50	50	58	70	84	102	116	126
50	50	52	58	69	86	101	120

$$*$$

0.1	0.1	0.1
0.1	0.2	0.1
0.1	0.1	0.1

$$=$$

69	95	116	125	129	132
68	92	110	120	126	132
66	86	104	114	124	132
62	78	94	108	120	129
57	69	83	98	112	124
53	60	71	85	100	114

 $f(x, y)$ $h(x, y)$ $g(x, y)$

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Key properties

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Linearity: $\text{filter}(f_1 + f_2) = \text{filter}(f_1) + \text{filter}(f_2)$

Shift invariance: same behavior regardless of pixel location: $\text{filter}(\text{shift}(f)) = \text{shift}(\text{filter}(f))$

Theoretical result: any linear shift-invariant operator can be represented as a convolution

Source: S. Lazebnik

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Properties in more detail

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Commutative: $a * b = b * a$

– Conceptually no difference between filter and signal

Associative: $a * (b * c) = (a * b) * c$

– Often apply several filters one after another: $((a * b_1) * b_2) * b_3$

– This is equivalent to applying one filter: $a * (b_1 * b_2 * b_3)$

Distributes over addition: $a * (b + c) = (a * b) + (a * c)$

Scalars factor out: $ka * b = a * kb = k(a * b)$

Identity: unit impulse $e = [\dots, 0, 0, 1, 0, 0, \dots]$,

$a * e = a$

Source: S. Lazebnik

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Integral image

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Precompute average



Get average values of any size by reading only 4 values!

Source: A. Tomrali

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Integral image



3	2	7	2	3
1	5	1	3	4
5	1	3	5	1
4	3	2	1	6
2	4	1	4	8

Original image

3	5	12	14	17
4	11	19	24	31
9	17	28	38	46
13	24	37	48	62
15	30	44	59	81

Integral image

$$s(i, j) = \sum_{k=0}^i \sum_{l=0}^j f(k, l)$$

Recursive:

$$s(i, j) = s(i-1, j) + s(i, j-1) - s(i-1, j-1) + f(i, j)$$

$$17+19-11+3 = 28$$

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Integral image



3	2	7	2	3
1	5	1	3	4
5	1	3	5	1
4	3	2	1	6
2	4	1	4	8

Original image

3	5	12	14	17
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13	24	37	48	62
15	30	44	59	81

Integral image

3	5	12	14	17
4	11	19	24	31
9	17	28	38	46
13	24	37	48	62
15	30	44	59	81

Integral image

4 values independent of size:

$$s(i_1, j_1) - s(i_1, j_0 - 1) - s(i_0 - 1, j_1) + s(i_0 - 1, j_0 - 1)$$

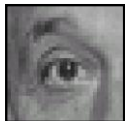
$$48-13-14+3 = 24 = 5+1+3+1+3+5+3+2+1$$

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Practice with linear filters



Original

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 0 \end{bmatrix} - \frac{1}{9} \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

(Note that filter sums to 1)

?

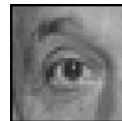
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Source: D. Lowe

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Practice with linear filters



Original

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 0 \end{bmatrix} - \frac{1}{9} \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

**Sharpening filter**

- Accentuates differences with local average

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Source: D. Lowe

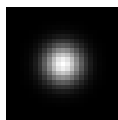
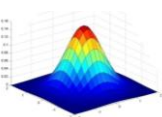
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Gaussian Kernel



Constant factor at front makes volume sum to 1 (can be ignored when computing the filter values, as we should renormalize weights to sum to 1 in any case)

$$G_{\sigma} = \frac{1}{2\pi\sigma^2} e^{-\frac{(x^2+y^2)}{2\sigma^2}}$$



0.003	0.013	0.022	0.013	0.003
0.013	0.059	0.097	0.059	0.013
0.022	0.097	0.159	0.097	0.022
0.013	0.059	0.097	0.059	0.013
0.003	0.013	0.022	0.013	0.003

5 x 5, $\sigma = 1$

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Source: C. Rasmussen

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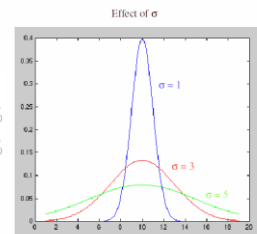
Choosing kernel width

Rule of thumb: set filter half-width to about 3σ

$$P(\mu - \sigma < X \leq \mu + \sigma) \approx 68\%$$

$$P(\mu - 2\sigma < X \leq \mu + 2\sigma) \approx 95.5\%$$

$$P(\mu - 3\sigma < X \leq \mu + 3\sigma) \approx 99.7\%$$



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Gaussian filters



Remove "high-frequency" components from the image (low-pass filter)

Convolution with self is another Gaussian

- So can smooth with small- σ kernel, repeat, and get same result as larger- σ kernel would have
- Convoluting two times with Gaussian kernel with std. σ is same as convoluting once with kernel with std. dev. $\sigma\sqrt{2}$

Separable kernel

- Factors into product of two 1D Gaussians

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Source: K. Grauman

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Why is separability useful?



What is the complexity of filtering an $n \times n$ image with an $m \times m$ kernel?

– $O(n^2 m^2)$

What if the kernel is separable?

– $O(n^2 m)$

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Source: S. Lazebnik

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Is a kernel separable?



- Kernel as matrix:

$$K = \sum_i \sigma_i u_i u_i^T$$

- Singular Value Decomposition (SVD)
- If only the first singular value σ_0 is non-zero, kernel is separable
- Vertical and horizontal kernels:

$$\sqrt{\sigma_0} u_0 \text{ and } \sqrt{\sigma_0} v_0^T$$

- Approximation: $K \approx \sigma_0 u_0 (v_0)^T$

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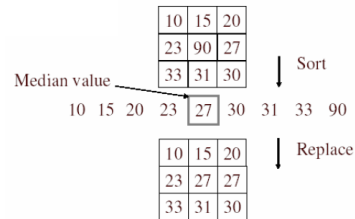
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Alternative idea: Median filtering



A median filter operates over a window by selecting the median intensity in the window



- Is median filtering linear?

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Source: K. Grauman

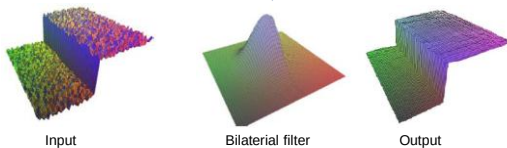
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Bilateral filter



Filter is data dependent

$$g(i, j) = \frac{\sum_{k,l} f(k, l) w(i, j, k, l)}{\sum_{k,l} w(i, j, k, l)}$$



$$w(i, j, k, l) = \exp \left(-\frac{(i-k)^2 + (j-l)^2}{2\sigma_s^2} - \frac{\|f(i, j) - f(k, l)\|^2}{2\sigma_r^2} \right)$$

Slow, but methods for approximation exist

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Linear filtering

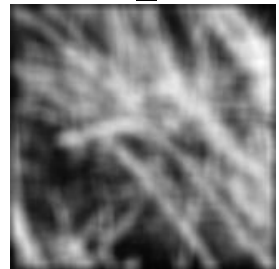
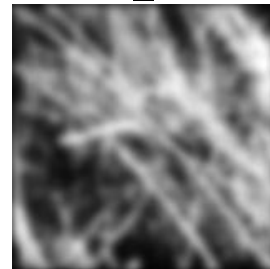


Gauss vs. box filter

Gaussian



Box filter



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A sum of sines

Our building block:

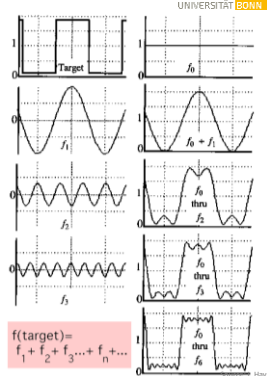
$$A \sin(\omega x + \phi)$$

A: Amplitude

ω : angular frequency

ϕ : phase

Add enough of them to get any signal $f(x)$ you want!



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Fourier Transform

Fourier transform stores the magnitude and phase at each frequency

- Magnitude encodes how much signal there is at a particular frequency
- Phase encodes spatial information (indirectly)
- For mathematical convenience, this is often notated in terms of real and complex numbers

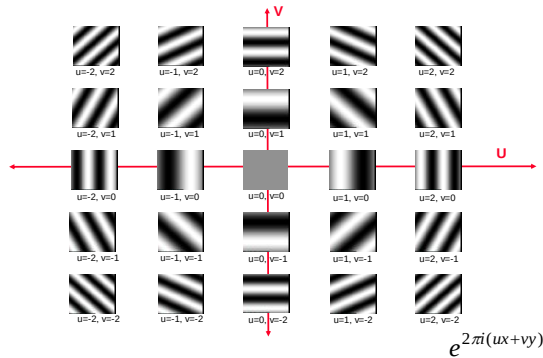
$$\text{Amplitude: } A = \pm \sqrt{R(\omega)^2 + I(\omega)^2} \quad \text{Phase: } \phi = \tan^{-1} \frac{I(\omega)}{R(\omega)}$$

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Source: J. Hays

The 2D Basis Functions

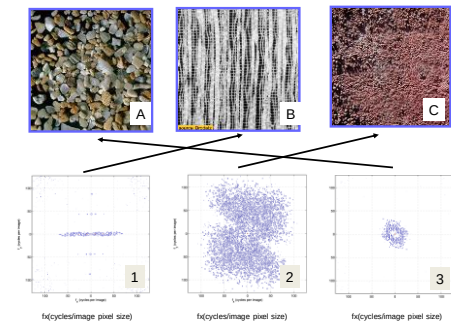


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Fourier Amplitude Spectrum



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Source: D. Hays

The Convolution Theorem

- The Fourier transform of the convolution of two functions is the product of their Fourier transforms

$$F[g * h] = F[g]F[h]$$

- The inverse Fourier transform of the product of two Fourier transforms is the convolution of the two inverse Fourier transforms

$$f[gh] = f[g] * f[h]$$

- **Convolution** in spatial domain is equivalent to **multiplication** in frequency domain!

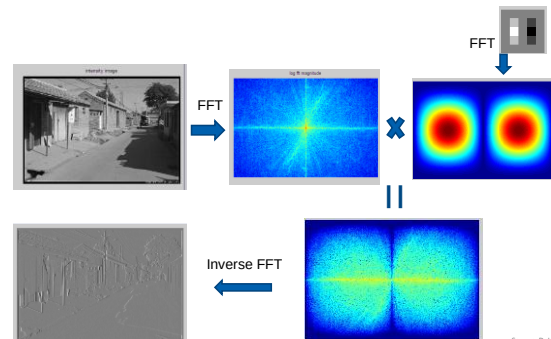
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Source: J. Hays

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Filtering in frequency domain



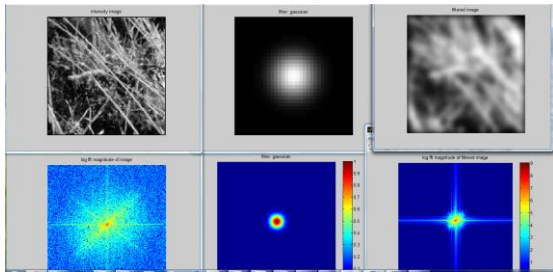
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Source: D. Hays

Gaussian

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Source: J. Hays

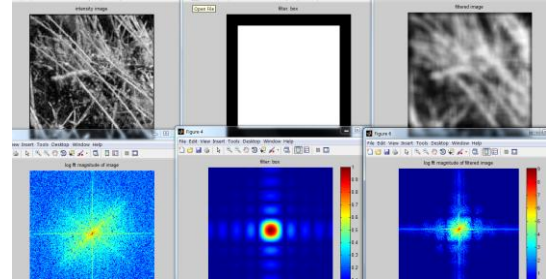
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Box filter

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Source: J. Hays

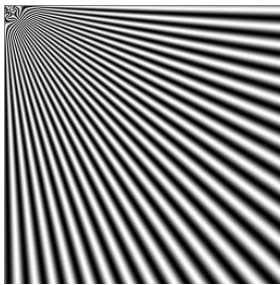
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Aliasing

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Source: S. Narasimhan

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Sampling Theorem

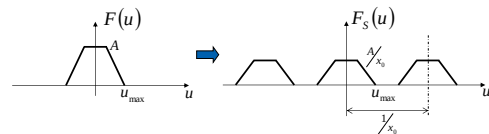
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Sampled function:

$$f_s(x) = f(x)s(x) = f(x) \sum_{n=-\infty}^{\infty} \delta(x - nx_0)$$

Sampling frequency $\frac{1}{x_0}$

$$F_s(u) = F(u) * S(u) = F(u) * \frac{1}{x_0} \sum_{n=-\infty}^{\infty} \delta\left(u - \frac{n}{x_0}\right)$$



$$\text{Only if } u_{\max} \leq \frac{1}{2x_0}$$

Source: S. Narasimhan

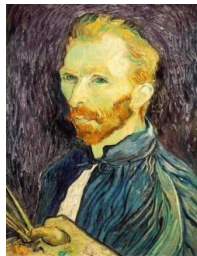
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Sub-Sampling with Gaussian Pre-Filtering

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Gaussian 1/2



G 1/4



G 1/8

- Solution: filter the image, *then* subsample
- Filter size should double for each 1/2 size reduction.

Source: S. Narasimhan

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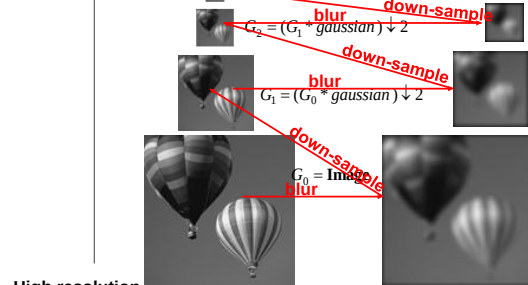
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Gaussian pyramid

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Low resolution

High resolution



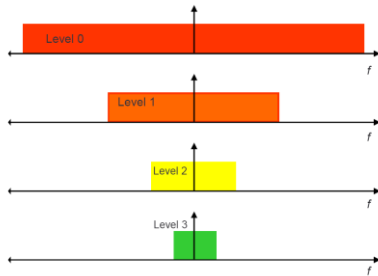
Source: S. Narasimhan

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Frequencies

Gaussian Pyramid
Frequency Composition

Source: S. Narasimhan

Sharpening

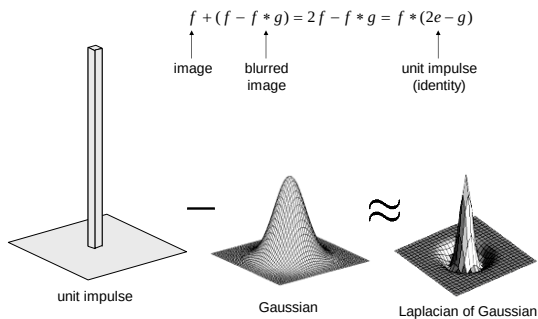
What does blurring take away?



Let's add it back:

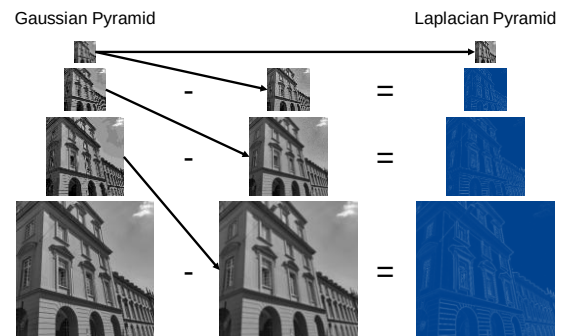


Laplacian of Gaussian

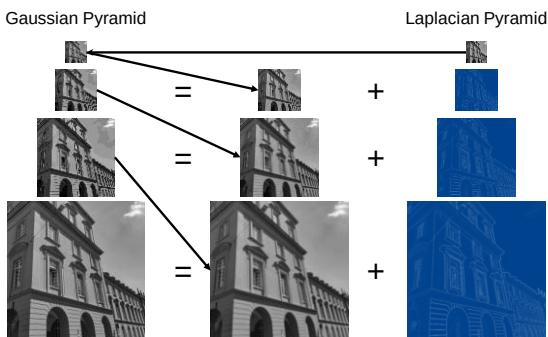


Source: S. Landin

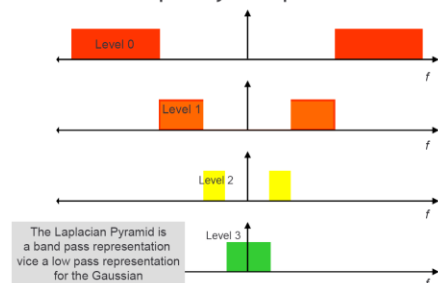
Laplacian pyramid



Laplacian pyramid



Frequencies

Laplacian Pyramid
Frequency Composition

The Laplacian Pyramid is a band pass representation vice a low pass representation for the Gaussian

Source: S. Narasimhan

Pyramid Blending

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Source: A. Effros

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Finite difference filters

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Other approximations of derivative filters exist:

$$\text{Prewitt: } M_x = \begin{bmatrix} -1 & 0 & 1 \\ -1 & 0 & 1 \\ -1 & 0 & 1 \end{bmatrix} ; M_y = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 0 & 0 \\ -1 & -1 & -1 \end{bmatrix}$$

$$\text{Sobel: } M_x = \begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix} ; M_y = \begin{bmatrix} 1 & 2 & 1 \\ 0 & 0 & 0 \\ -1 & -2 & -1 \end{bmatrix}$$

$$\text{Roberts: } M_x = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} ; M_y = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

Source: K. Grauman

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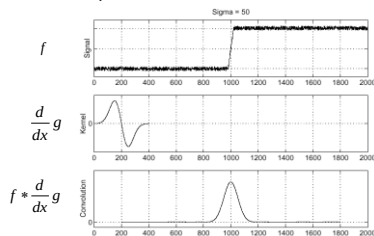
Derivative theorem of convolution

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Differentiation is convolution, and convolution is associative:

$$\frac{d}{dx}(f * g) = f * \frac{d}{dx}g$$

This saves us one operation:



Source: S. Seitz

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The Canny edge detector

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original image

high threshold
(strong edges)low threshold
(weak edges)

hysteresis threshold

Source: L. Felkel

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Canny edge detector

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1. Filter image with derivative of Gaussian
2. Find magnitude and orientation of gradient
3. **Non-maximum suppression:**
 - Thin wide "ridges" down to single pixel width
4. **Linking and thresholding (hysteresis):**
 - Define two thresholds: low and high
 - Use the high threshold to start edge curves and the low threshold to continue them

J. Canny, *A Computational Approach To Edge Detection*, IEEE Trans. Pattern Analysis and Machine Intelligence, 8:679-714, 1986.

Source: S. Lazebnik

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Distance transform

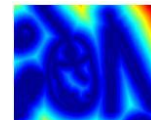
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original



edges



distance transform

Value at (x,y) tells how far that position is from the nearest edge point (or other binary image structure)

Source: K. Grauman

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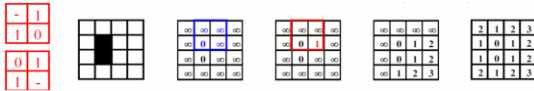
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Distance Transform (2D)



- 2D case analogous to 1D
 - Initialization
 - Forward and backward pass
 - Fwd pass finds closest above and to left
 - Bwd pass finds closest below and to right



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Chamfer distance



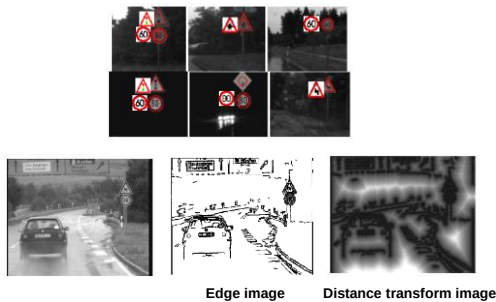
- Average distance to nearest feature

$$D_{\text{chamfer}}(T, I) \equiv \frac{1}{|T|} \sum_{t \in T} d_I(t)$$

- T: template shape $\rightarrow$ a set of points
- I: image to search $\rightarrow$ a set of points
- $d(t)$: min distance for point t to some point in I

12/22/2045 Jürgen Gall - Institute of Computer Science III - Computer Vision Group Source: K. Grauman

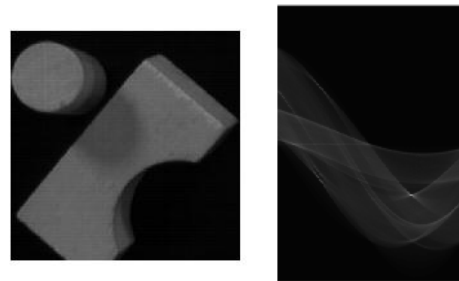
Chamfer distance



Edge image Distance transform image

11/21/2035 Jürgen Gall - Institute of Computer Science III - Computer Vision Group Source: D. Gavrila

Hough transform



12/22/2015 Jürgen Gall - Institute of Computer Science III - Computer Vision Group Source: K. Grauman

Hough transform algorithm

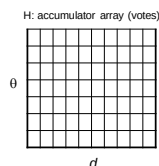


Using the polar parameterization:

$$x \cos \theta - y \sin \theta = d$$

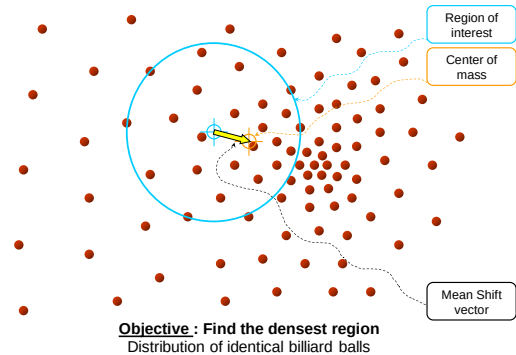
Basic Hough transform algorithm

- Initialize $H[d, \theta] = 0$
- for each edge point $I[x, y]$ in the image
for $\theta = 0$ to 180 // some quantization
 $d = x \cos \theta - y \sin \theta$
 $H[d, \theta] += 1$
- Find the value(s) of (d, θ) where $H[d, \theta]$ is maximum
- The detected line in the image is given by $d = x \cos \theta - y \sin \theta$



11/22/2035 Jürgen Gall - Institute of Computer Science III - Computer Vision Group Source: S. Seitz

Mean Shift



Objective: Find the densest region
Distribution of identical billiard balls

12/22/2015 Jürgen Gall - Institute of Computer Science III - Computer Vision Group Source: Y. Ukita and B. Sarel

Mean shift

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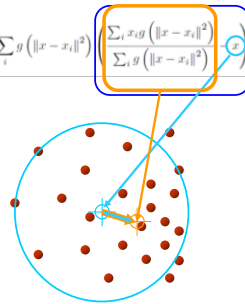
$$\nabla P(x) = \frac{C}{n} \sum_{i=1}^n \nabla k_i = \frac{2C}{N} \sum_i g \left(\|x - x_i\|^2 \right) \left(\frac{\sum_i x_i g \left(\|x - x_i\|^2 \right)}{\sum_i g \left(\|x - x_i\|^2 \right)} - x \right)$$

Simple Mean Shift procedure:

- Compute mean shift vector

$$m(x) = \left[\frac{\sum_{i=1}^n x_i g \left(\frac{\|x - x_i\|^2}{h} \right)}{\sum_{i=1}^n g \left(\frac{\|x - x_i\|^2}{h} \right)} - x \right]$$

- Translate the Kernel window by $m(x)$



Source: Y. Uharantz and B. Sarel

12112005

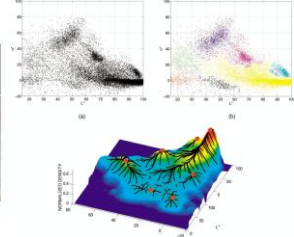
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Mean shift clustering/segmentation

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- Find features (color, gradients, texture, etc)
- Initialize windows at individual feature points
- Perform mean shift for each window until convergence
- Merge windows that end up near the same "peak" or mode



Source: K. Grauman

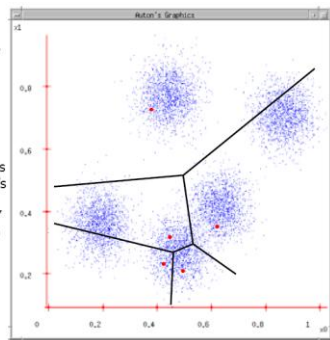
10200005

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K-means

1. Ask user how many clusters they'd like. (e.g. $k=5$)
2. Randomly guess k cluster Center locations
3. Each datapoint finds out which Center it's closest to. (Thus each Center "owns" a set of datapoints)



Source: A. Moore

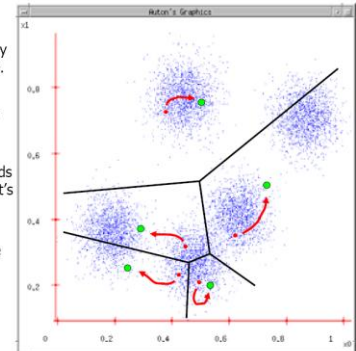
10102005

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K-means

1. Ask user how many clusters they'd like. (e.g. $k=5$)
2. Randomly guess k cluster Center locations
3. Each datapoint finds out which Center it's closest to.
4. Each Center finds the centroid of the points it owns



Source: A. Moore

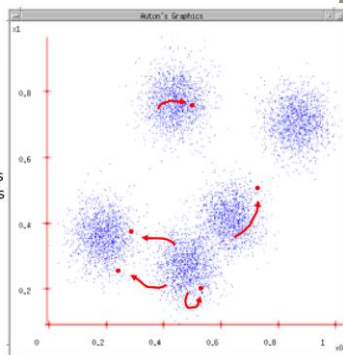
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K-means

1. Ask user how many clusters they'd like. (e.g. $k=5$)
2. Randomly guess k cluster Center locations
3. Each datapoint finds out which Center it's closest to.
4. Each Center finds the centroid of the points it owns...
5. ...and jumps there
6. ...Repeat until terminated!



Source: A. Moore

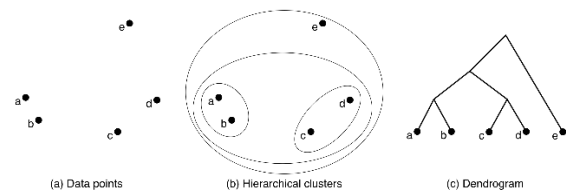
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Agglomerative Clustering

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Source: R. Akella

10202005

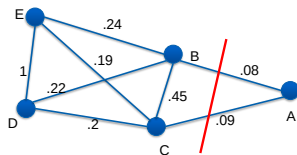
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Spectral Clustering Example



- Minimum Cut



$$\text{Cut}(\text{BCDE}, \text{A}) = 0.08 + 0.09 = 0.17$$

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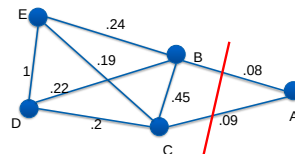
81

Spectral Clustering Example



- Normalized Minimum Cut

$$\text{NormCut}(C_1, C_2) = \frac{\text{Cut}(C_1, C_2)}{\text{Vol}(C_1)} + \frac{\text{Cut}(C_1, C_2)}{\text{Vol}(C_2)}$$



$$\text{Cut}(\text{BCDE}, \text{A}) = 0.17 / (0.99 + 0.93 + 1.42 + 1.43) + 0.17 / 0.17 = 1.036$$

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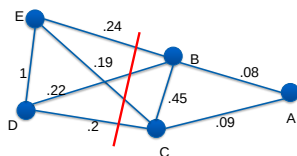
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Spectral Clustering Example



- Normalized Minimum Cut

$$\text{NormCut}(C_1, C_2) = \frac{\text{Cut}(C_1, C_2)}{\text{Vol}(C_1)} + \frac{\text{Cut}(C_1, C_2)}{\text{Vol}(C_2)}$$



$$\begin{aligned} \text{Cut}(\text{BCDE}, \text{A}) &= 0.17 / (0.99 + 0.93 + 1.42 + 1.43) + 0.17 / 0.17 = 1.036 \\ \text{Cut}(\text{DE}, \text{ABC}) &= 0.85 / (1.42 + 1.43) + 0.85 / (0.17 + 0.99 + 0.93) = 0.705 \end{aligned}$$

11/01/2005

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Spectral Clustering



- Construct an **affinity** matrix



W

	A	B	C	D
A	0	.2	.2	0
B	.2	0	.3	0
C	.2	.3	0	.1
D	0	0	.1	0

- Graph Laplacian

$$L = D - W$$

	A	B	C	D
A	.4	-.2	-.2	0
B	-.2	.5	-.3	0
C	-.2	-.3	.6	-.1
D	0	0	-.1	.1

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Spectral Clustering



Normalized Minimum Cut (Relaxed):

$$\min_x \text{Ncut}(x) = \min_y \frac{y^T (D - W) y}{y^T D y}$$

Generalized eigenvalue problem:

$$(D - W)y = \lambda D y$$

Equal eigenvalue problem:

$$D^{-\frac{1}{2}}(D - W)D^{-\frac{1}{2}}z = \lambda z \quad z = D^{\frac{1}{2}}y$$

Eigenvector corresponding to the second smallest eigenvalue

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More than two clusters, Laplacian eigenmaps

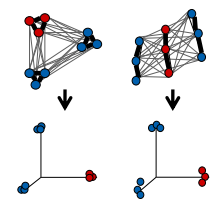


The eigenvector to the third smallest eigenvalue indicates an alternative partitioning of the graph

In the general case we compute the k smallest eigenvalues

The corresponding eigenvectors span a subspace

- $Y = (y_1, y_2, \dots, y_k) \in \mathbb{R}^{N \times k}$ matrix
- Each data point $1, 2, \dots, N$ maps to a k-dimensional vector (i-th row of Y)
- Mapping is called **Laplacian eigenmap**

Standard clustering techniques (k-means) to convert the real-valued vectors into integer labels
→ **spectral clustering**

Mappings from a graph to its eigenspace, clustering in the eigenspace

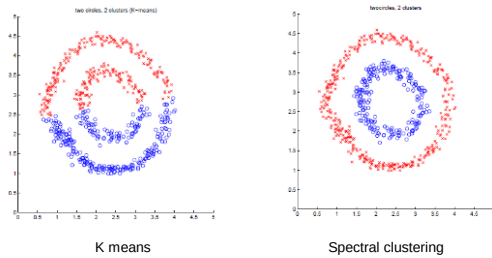
[Ulrike von Luxburg, A Tutorial on Spectral Clustering, Statistics and Computing 2007]

22.12.2005

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K means vs. spectral clustering



A. Ng et al. On spectral clustering: Analysis and an algorithm. NIPS 2001

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Deformable contours



a.k.a. active contours, snakes

Given: initial contour (model) near desired object

Goal: evolve the contour to fit exact object boundary



Main idea: elastic band is iteratively adjusted so as to

- be near image positions with high gradients, **and**
- satisfy shape "preferences" or contour priors

[Snakes: Active contour models, Kass, Witkin, & Terzopoulos, ICCV1987]

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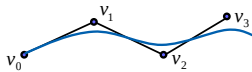
Source: Y. Boykov

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B-spline snakes



Explicit representation of contour: $C(s) = \sum_{i=0}^{N-1} v_i \cdot B_i(s)$



Smooth (differentiable) curve passes close to the poly-line joining the knot (control) points

B-spline snakes - **parametric** representation of continuous contours via finite number of parameters:

$$\{v_0, v_1, v_2, \dots, v_{N-1}\}$$

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Source: Y. Boykov

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Energy function



The total energy (cost) of the current snake is defined as:

$$E_{total} = E_{internal} + E_{external}$$



Internal energy: encourage *prior* shape preferences: e.g., smoothness, elasticity, particular known shape.

External energy ("image" energy): encourage contour to fit on places where image structures exist, e.g., edges.

A good fit between the current deformable contour and the target shape in the image will yield a **low** value for this cost function.

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Source: K. Grauman

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Total energy: function of the weights



$$E_{total} = E_{internal} + \gamma E_{external}$$

$$E_{external} = - \sum_{i=0}^{n-1} |G_x(x_i, y_i)|^2 + |G_y(x_i, y_i)|^2$$

$$E_{internal} = \sum_{i=0}^{n-1} \left(\alpha (\bar{d} - \|v_{i+1} - v_i\|)^2 + \beta \|v_{i+1} - 2v_i + v_{i-1}\|^2 \right)$$

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Source: K. Grauman

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The Snakes model (continuous)



Kass, Witkin, and Terzopoulos proposed the following energy functional:

$$E(C) = \underbrace{- \int_0^1 |\nabla I(C(s))|^2 ds}_{\text{External energy (data)}} + \alpha \underbrace{\int_0^1 |C_s(s)|^2 ds + \beta \int_0^1 |C_{ss}(s)|^2 ds}_{\text{Internal energy (prior)}}$$

$C: [0, 1] \rightarrow \Omega$ is a parametric contour. C_s and C_{ss} denote the first and second derivative of this contour.

The first term is called **external energy**, since it depends on the (external) input image. The second two terms are called **internal energy**, since they are inherent to the model and independent of the data.

Minimizing the external energy drives the contour to go along maxima of the gradient.

[M. Kass, A. Witkin, D. Terzopoulos. Snakes: Active contour models. IJCV 1987]

22.12.2015

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Source: T. Brox

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Energy minimization



- Several algorithms have been proposed to fit deformable contours.
- We'll look at two:
 - Greedy search
 - Dynamic programming (for 2d snakes)
 - Variational optimization

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Dynamic programming



- General minimization problem with unary U_n and pairwise terms P_n :

$$\operatorname{argmin}_{w_1, \dots, w_N} \left[\sum_{n=1}^N U_n(w_n) + \sum_{n=1}^{N-1} P_n(w_n, w_{n+1}) \right]$$

- Snakes with control points w_n :

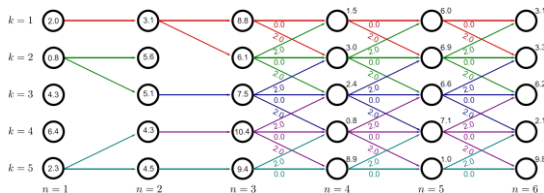
$$U_n(w_n) = -\|G(w_n)\|^2$$

$$P_n(w_n, w_{n+1}) = \alpha \|w_{n+1} - w_n\|^2$$

- States of $w_n=k$ are pixels in neighborhood: 

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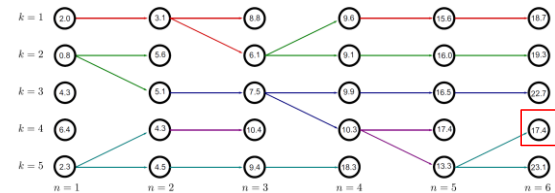
Worked example



Work through the graph, computing the minimum cost to reach each node

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Worked example



Find the minimum possible cost to reach the final column

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Worked example



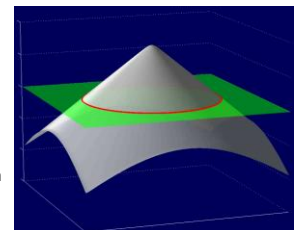
Trace back the route that we arrived here by – this is the minimum configuration

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Level set representation of contours



- Introduce an **embedding function** $\phi : \Omega \rightarrow \mathbb{R}$
- The zero-level line represents the contour (can be any other level)
 $C = \{x \in \Omega \mid \phi(x) = 0\}$
- For evolving C evolve ϕ
- No regridding necessary
- Allows for topological changes
- Can be applied in any dimension
- Represents the contour *and* the enclosed region



10/02/2005 Jürgen Gall - Institute of Computer Science III - Computer Vision Group Source: D. Cremers

Geodesic active contours

Caselles et al. **Geodesic Active Contours**. IJCV 1997 introduced two improvements of the snake model.

1. A level set representation of the contour

2. A variation of the model, called the **geodesic active contour**:

$$E(C) = \int_0^1 w(|\nabla I(C(s))|^2) |C_s(s)| ds = \int_C w(|\nabla I|^2) ds$$

It consists of a single term that penalizes the **geodesic length** of the contour using a metric induced by the image gradient.

For comparison: measuring the Euclidean length of a contour reads:

$$E(C) = \int_0^1 |C_s(s)| ds = \int_C ds$$

→ Combines the internal and external energy of the Snake model

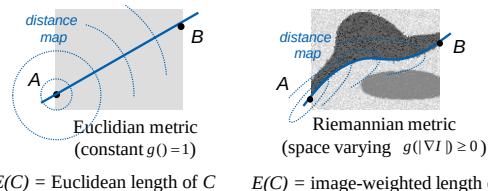
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Geodesic active contours

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- Energy $E(C) = \int_A^B g(|\nabla I_{C(s)}|) ds$ = “weighted length” of C
- Geodesic**: Shortest curve between two points (wrt. E).



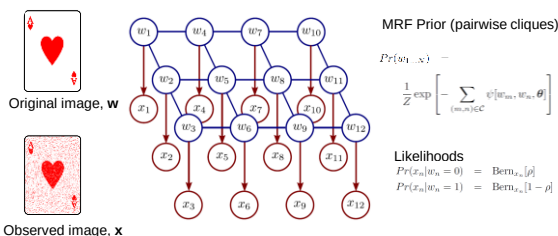
$E(C)$ = Euclidean length of C $E(C)$ = image-weighted length of C

Source: Y. Boykov

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Denoising with MRFs

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Observed image, x

$$\text{Inference: } Pr(w_1...w_N | x_1...x_N) = \frac{\prod_{n=1}^N Pr(x_n | w_n) Pr(w_1...w_N)}{Pr(x_1...x_N)}$$

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Graph Cuts Overview

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Graph cuts used to optimise this cost function:

$$\underset{w_1...w_N}{\operatorname{argmin}} \left[\sum_{n=1}^N U_n(w_n) + \sum_{(m,n) \in C} P_{mn}(w_m, w_n) \right]$$

Unary terms (compatibility of data with label y) **Pairwise terms** (compatibility of neighboring labels)

Approach:

Convert minimization into the form of a standard CS problem,

MAXIMUM FLOW or MINIMUM CUT ON A GRAPH

Polynomial-time methods for solving this problem are known

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General Pairwise costs

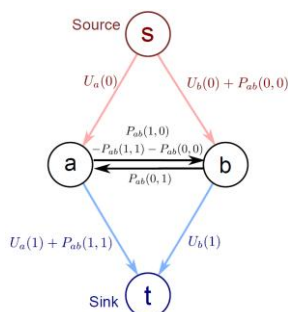
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$$P_{m,n}(0,0) = \theta_{00} \quad P_{m,n}(1,0) = \theta_{10} \\ P_{m,n}(0,1) = \theta_{01} \quad P_{m,n}(1,1) = \theta_{11}$$

Modify graph to

- Add $P(0,0)$ to edge s-b
 - Implies that solutions 0,0 and 1,0 also pay this cost
- Subtract $P(0,0)$ from edge b-a
 - Solution 1,0 has this cost removed again

Similar approach for $P(1,1)$



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Submodularity

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$$\theta_{10} + \theta_{01} - \theta_{11} - \theta_{00} \geq 0$$

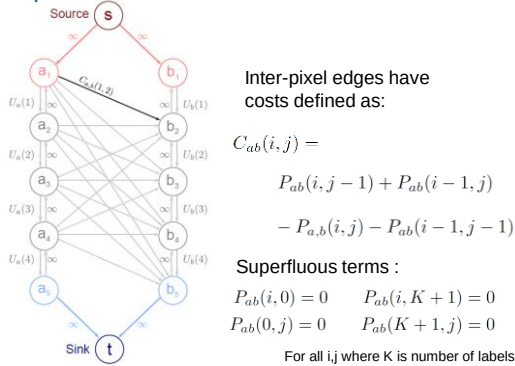
If this condition is obeyed, it is said that the problem is “submodular” and it can be solved in polynomial time.

If it is not obeyed then the problem is NP hard.

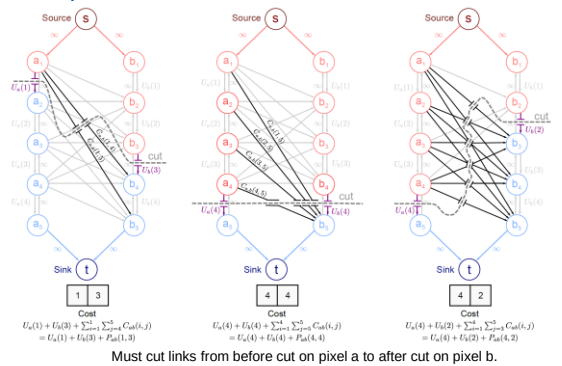
Usually it is not a problem as we tend to favour smooth solutions.

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Multiple Labels

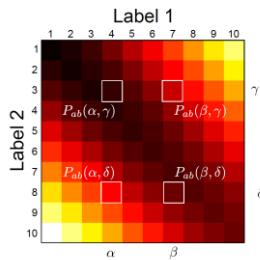


Example Cuts



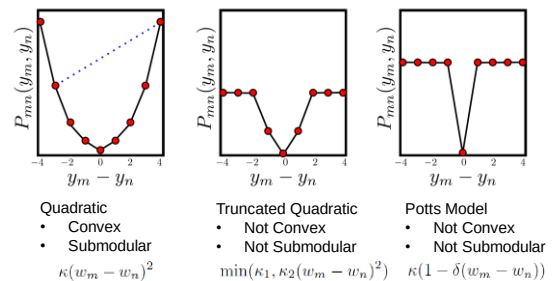
Submodularity

$$P_{ab}(\beta, \gamma) + P_{ab}(\alpha, \delta) - P_{ab}(\beta, \delta) - P_{ab}(\alpha, \gamma) \geq 0 \quad \beta > \alpha \text{ and } \delta > \gamma$$

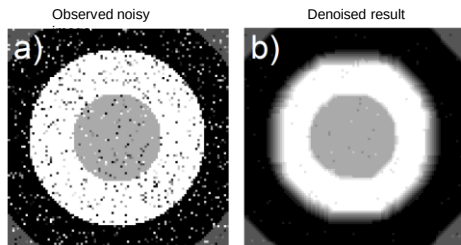


If not submodular, then the problem is NP hard.

Convex vs. non-convex costs



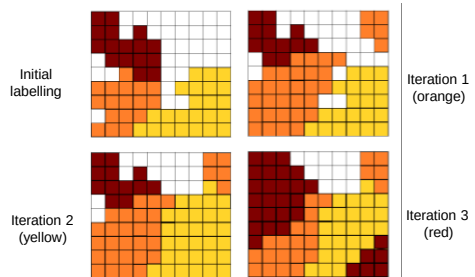
What is wrong with convex costs?



- Pay lower price for many small changes than one large one
- Result: blurring at large changes in intensity

Alpha Expansion Algorithm

- break multilabel problem into a series of binary problems
- at each iteration, pick label α and expand (retain original or change to α)



Alpha Expansion Ideas

- For every iteration
 - For every label
 - Expand label using optimal graph cut solution

Co-ordinate descent in label space.

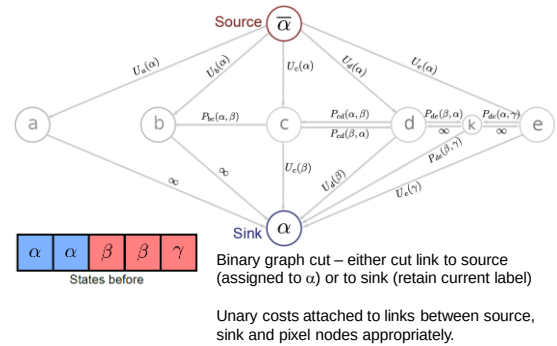
Each step optimal, but overall global maximum not guaranteed

Proved to be within a factor of 2 of global optimum.

Requires that pairwise costs form a metric:

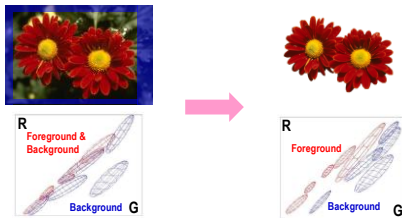
$$\begin{aligned} P(\alpha, \beta) &= 0 \iff \alpha = \beta \\ P(\alpha, \beta) &= P(\beta, \alpha) \geq 0 \\ P(\alpha, \beta) &\leq P(\alpha, \gamma) + P(\gamma, \beta) \end{aligned}$$

Alpha Expansion Construction



Grab Cut

- Loosely specify foreground region
- Iterated graph cut



Gaussian Mixture Model (typically 5-8 components)

Expectation Maximization

An algorithm specialized to fitting pdfs which are the marginalization of a joint distribution

$$\hat{\theta} = \operatorname{argmax}_{\theta} \left[\sum_{i=1}^I \log \left[\int Pr(\mathbf{x}_i, \mathbf{h}_i | \theta) d\mathbf{h}_i \right] \right]$$

Defines a lower bound on log likelihood and increases bound iteratively

$$\begin{aligned} \mathcal{B}[\{q_i(\mathbf{h}_i)\}, \theta] &= \sum_{i=1}^I \int q_i(\mathbf{h}_i) \log \left[\frac{Pr(\mathbf{x}_i, \mathbf{h}_i | \theta)}{q_i(\mathbf{h}_i)} \right] d\mathbf{h}_{1...I} \\ &\leq \sum_{i=1}^I \log \left[\int Pr(\mathbf{x}_i, \mathbf{h}_i | \theta) d\mathbf{h}_{1...I} \right] \end{aligned}$$

E-Step & M-Step

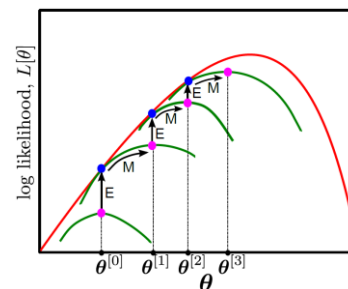
E-Step – Maximize bound w.r.t. distributions $q_i(\mathbf{h}_i)$

$$\hat{q}_i(\mathbf{h}_i) = Pr(\mathbf{h}_i | \mathbf{x}_i, \theta^{[t]}) = \frac{Pr(\mathbf{x}_i | \mathbf{h}_i, \theta^{[t]}) Pr(\mathbf{h}_i | \theta^{[t]})}{Pr(\mathbf{x}_i)}$$

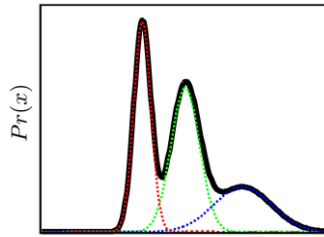
M-Step – Maximize bound w.r.t. parameters θ

$$\hat{\theta}^{[t+1]} = \operatorname{argmax}_{\theta} \left[\sum_{i=1}^I \int \hat{q}_i(\mathbf{h}_i) \log [Pr(\mathbf{x}_i, \mathbf{h}_i | \theta)] d\mathbf{h}_i \right]$$

E-Step & M-Step



Mixture of Gaussians (MoG)



$$Pr(\mathbf{x}|\boldsymbol{\theta}) = \sum_{k=1}^K \lambda_k \text{Norm}_{\mathbf{x}}[\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k]$$

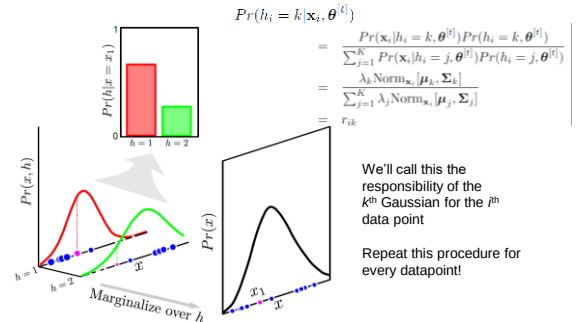
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Source: S. Prince

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E-Step



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M-Step



$$\hat{\boldsymbol{\theta}}^{[t+1]} = \underset{\boldsymbol{\theta}}{\text{argmax}} \left[\sum_{i=1}^I \sum_{k=1}^K \hat{q}_i(h_i = k) \log [Pr(\mathbf{x}_i, h_i = k | \boldsymbol{\theta})] \right]$$

$$= \underset{\boldsymbol{\theta}}{\text{argmax}} \left[\sum_{i=1}^I \sum_{k=1}^K r_{ik} \log [\lambda_k \text{Norm}_{\mathbf{x}_i}[\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k]] \right]$$

Take derivative, equate to zero and solve (Lagrange multipliers for λ)

$$\lambda_k^{[t+1]} = \frac{\sum_{i=1}^I r_{ik}}{\sum_{j=1}^K \sum_{i=1}^I r_{ij}}$$

$$\boldsymbol{\mu}_k^{[t+1]} = \frac{\sum_{i=1}^I r_{ik} \mathbf{x}_i}{\sum_{i=1}^I r_{ik}}$$

$$\boldsymbol{\Sigma}_k^{[t+1]} = \frac{\sum_{i=1}^I r_{ik} (\mathbf{x}_i - \boldsymbol{\mu}_k^{[t+1]})(\mathbf{x}_i - \boldsymbol{\mu}_k^{[t+1]})^T}{\sum_{i=1}^I r_{ik}}$$

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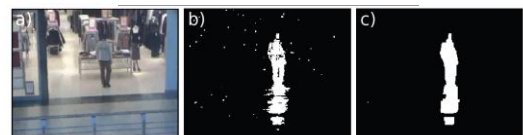
Background subtraction



Model background distribution of each pixel by GMM:

$$Pr(\mathbf{x}_n | w = 0) = \sum_i \lambda_{n,i} \text{Norm}_{\mathbf{x}_n}[\boldsymbol{\mu}_{n,i}, \boldsymbol{\Sigma}_{n,i}]$$

$$Pr(\mathbf{x}_n | w = 1) = \kappa,$$



Test image

Pixel-wise classification

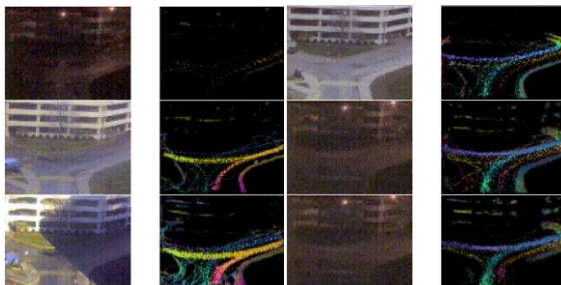
MRF

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Cars and pedestrians



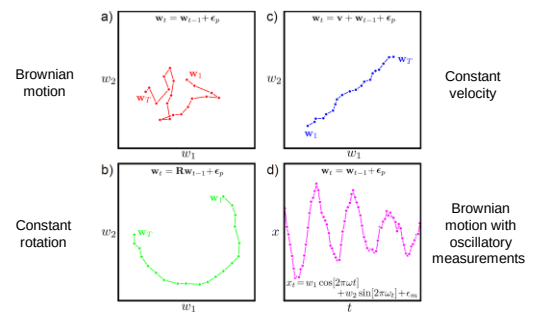
1 day

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Temporal Models



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Source: S. Prince

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Assumptions

- Only the immediate past matters (Markov)
 - the probability of the state at time t is conditionally independent of states at times $1 \dots t-2$ given the state at time $t-1$, i.e., $P(w_t | w_{t-1}, \dots, w_1) = P(w_t | w_{t-1})$
- Measurements depend on only the current state
 - the likelihood of the measurements at time t is conditionally independent of all of the other measurements and the states at times $1 \dots t-1$ given the state at time t , $P(x_t | w_t, \dots, w_1, x_{t-1}, \dots, x_1) = P(x_t | w_t)$

Source: S. Prince

Recursive Estimation

Time t

$$\begin{aligned}
 Pr(w_t | x_t, x_{1 \dots t-1}) &= \frac{Pr(x_t | w_t, x_{1 \dots t-1}) Pr(w_t | x_{1 \dots t-1})}{Pr(x_t | x_{1 \dots t-1})} \\
 &= \frac{Pr(x_t | w_t) Pr(w_t | x_{1 \dots t-1})}{\int Pr(x_t | w_t) Pr(w_t | x_{1 \dots t-1}) dw_t} \\
 &= \frac{Pr(x_t | w_t) Pr(w_t | x_{1 \dots t-1})}{\int Pr(x_t | w_t) Pr(w_t | x_{1 \dots t-1}) dw_t}
 \end{aligned}$$

Measurement model

Prediction from temporal model

Source: S. Prince

Summary

Alternate between:

Temporal Evolution

Temporal model

$$Pr(w_t | x_{1 \dots t-1}) = \int Pr(w_t | w_{t-1}) Pr(w_{t-1} | x_{1 \dots t-1}) dw_{t-1}$$

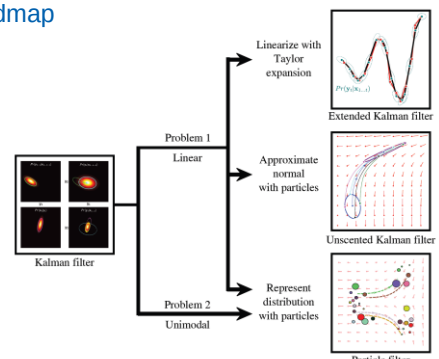
Measurement Update

Measurement model

$$Pr(w_t | x_{1 \dots t}) = \frac{Pr(x_t | w_t) Pr(w_t | x_{1 \dots t-1})}{\int Pr(x_t | w_t) Pr(w_t | x_{1 \dots t-1}) dw_t}$$

Source: S. Prince

Roadmap



Source: S. Prince

Kalman Filter Definition

Time evolution equation

$$Pr(w_t | w_{t-1}) = \text{Norm}_{w_t}[\mu_p + \Psi w_{t-1}, \Sigma_p]$$

State transition matrix

Additive Gaussian noise

Measurement equation

$$Pr(x_t | w_t) = \text{Norm}_{x_t}[\mu_m + \Phi w_t, \Sigma_m]$$

Relates state and measurement

Additive Gaussian noise

Source: S. Prince

Kalman Filter Summary

Time evolution equation

$$Pr(w_t | w_{t-1}) = \text{Norm}_{w_t}[\mu_p + \Psi w_{t-1}, \Sigma_p]$$

Measurement equation

$$Pr(x_t | w_t) = \text{Norm}_{x_t}[\mu_m + \Phi w_t, \Sigma_m]$$

Inference

$$\begin{aligned}
 \text{State Prediction: } \mu_+ &= \mu_p + \Psi \mu_{t-1} \\
 \text{Covariance Prediction: } \Sigma_+ &= \Sigma_p + \Psi \Sigma_{t-1} \Psi^T \\
 \text{State Update: } \mu_t &= \mu_+ + K(x_t - \mu_m - \Phi \mu_+) \\
 \text{Covariance Update: } \Sigma_t &= (I - K\Phi) \Sigma_+,
 \end{aligned}$$

$$K = \Sigma_+ \Phi^T (\Sigma_m + \Phi \Sigma_+ \Phi^T)^{-1}$$

Source: S. Prince

Fixed lag smoother



State evolution equation

$$\begin{bmatrix} \mathbf{w}_t^{[1]} \\ \mathbf{w}_t^{[2]} \\ \vdots \\ \mathbf{w}_t^{[r]} \end{bmatrix} = \begin{bmatrix} \Psi & 0 & \dots & 0 & 0 \\ \mathbf{I} & 0 & \dots & 0 & 0 \\ 0 & \mathbf{I} & \dots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \dots & \mathbf{I} & 0 \end{bmatrix} \begin{bmatrix} \mathbf{w}_{t-1}^{[1]} \\ \mathbf{w}_{t-1}^{[2]} \\ \vdots \\ \mathbf{w}_{t-1}^{[r]} \end{bmatrix} + \begin{bmatrix} \epsilon_p \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

Estimate delayed by τ →

Measurement equation

$$\mathbf{x}_t = \begin{bmatrix} \Phi & 0 & 0 & \dots & 0 \end{bmatrix} \begin{bmatrix} \mathbf{w}_t^{[1]} \\ \mathbf{w}_t^{[2]} \\ \vdots \\ \mathbf{w}_t^{[r]} \end{bmatrix} + \epsilon_m$$

Source: S. Prince

Fixed interval smoothing



Having μ_t , Σ_t , $\mu_{+|t}$ and $\Sigma_{+|t}$ from forward pass where $\Pr(\mathbf{w}_{t+1}|\mathbf{x}_{1:t}) = \text{Norm}(\mu_{+|t}, \Sigma_{+|t})$

Backward set of recursions

$$\begin{aligned} \mu_{t|T} &= \mu_t + \mathbf{C}_t(\mu_{t+1|T} - \mu_{+|t}) \\ \Sigma_{t|T} &= \Sigma_t + \mathbf{C}_t(\Sigma_{t+1|T} - \Sigma_{+|t})\mathbf{C}_t^T \end{aligned}$$

where

$$\mathbf{C}_t = \Sigma_t \Psi^T \Sigma_{+|t}^{-1}$$

Source: S. Prince

Learning parameters



Parameters of equations:

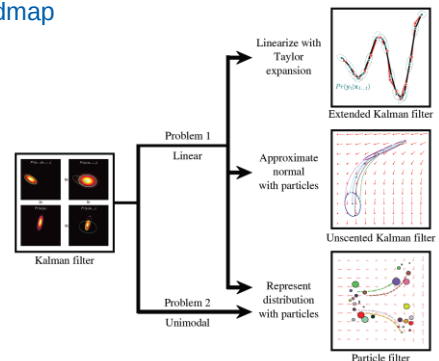
$$\begin{aligned} w_t &= \Psi w_{t-1} + \epsilon_p \\ x_t &= \Phi w_t + \epsilon_m \end{aligned} \quad \theta = \{\Psi, \Sigma_p, \Phi, \Sigma_m, \mu_0, \Sigma_0\}$$

Expectation-Maximation

- E step:
 - Fixed interval smoothing to get (w_t)
- M step:
 - Maximize log-likelihood to get θ

Source: S. Prince

Roadmap



Source: S. Prince

Extended Kalman Filter



Allows non-linear measurement and temporal equations

$$\begin{aligned} \mathbf{w}_t &= \mathbf{f}[\mathbf{w}_{t-1}, \epsilon_p] \\ \mathbf{x}_t &= \mathbf{g}[\mathbf{w}_t, \epsilon_m] \end{aligned}$$

Key idea: take Taylor expansion and treat as locally linear

Source: S. Prince

Unscented Kalman Filter



Approximate with particles where D_w is dimensionality of state:

$$\begin{aligned} \Pr(\mathbf{w}_{t-1}|\mathbf{x}_{1:t-1}) &= \text{Norm}_{\mathbf{w}_{t-1}}[\mu_{t-1}, \Sigma_{t-1}] \\ &\approx \sum_{j=0}^{2D_w} a_j \delta[\mathbf{w}_{t-1} - \hat{\mathbf{w}}^{[j]}] \end{aligned}$$

Choose so that

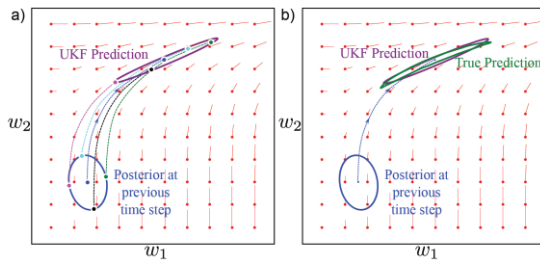
$$\begin{aligned} \mu_{t-1} &= \sum_{j=0}^{2D_w} a_j \hat{\mathbf{w}}^{[j]} \\ \Sigma_{t-1} &= \sum_{j=0}^{2D_w} a_j (\hat{\mathbf{w}}^{[j]} - \mu_{t-1})(\hat{\mathbf{w}}^{[j]} - \mu_{t-1})^T \end{aligned}$$

weights $\sum_{j=0}^{2D_w} a_j = 1$ particles or sigma points $\delta(x) = \begin{cases} 1 & x = 0 \\ 0 & \text{otherwise} \end{cases}$

Source: S. Prince

Unscented Kalman Filter

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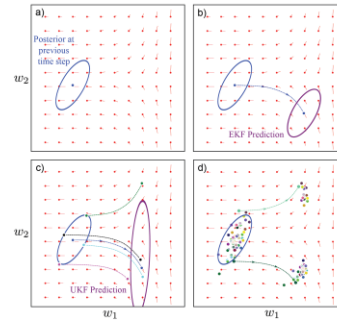
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Source: S. Prince

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Problems with UKF

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Particle filters

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Key idea:

- Represent probability distribution as a set of weighted particles

$$Pr(\mathbf{w}_{t-1} | \mathbf{x}_{1:t-1}) = \sum_{j=1}^J a_j \delta[\mathbf{w}_{t-1} - \bar{\mathbf{w}}_{t-1}^{[j]}]$$

Advantages and disadvantages:

- + Can represent non-Gaussian multimodal densities
- + No need for data association
- Expensive

N. Gordon et al. *Novel approach to nonlinear/non-Gaussian Bayesian state estimation*, 1993
M. Isard and A. Blake. *Contour tracking by stochastic propagation of conditional density*, ECCV 1996

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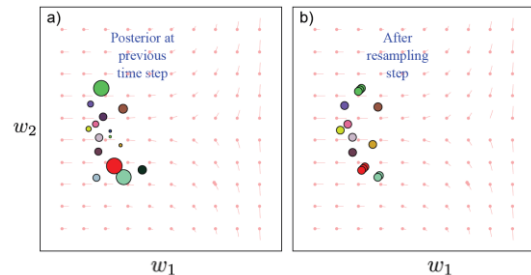
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Particle filter

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Stage 1: Resample from weighted particles according to their weight to get unweighted particles



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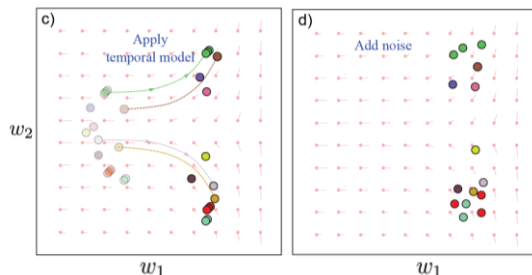
Source: S. Prince

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Stage 2: Pass unweighted samples through temporal model and add noise



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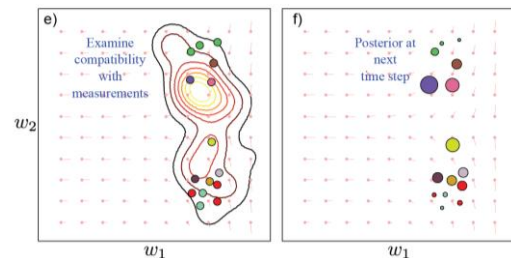
Source: S. Prince

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Stage 3: Weight samples by measurement density



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Source: S. Prince

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Generic particle filter



1. Initialisation

- $t \leftarrow 0$
- For $i = 1, \dots, n$, sample $x_{0,0}^{(i)}$ from η_0

2. Prediction

Temporal model

- For $i = 1, \dots, n$, sample $\bar{x}_{t+1,0}^{(i)}$ from $K_t(x_{t,0}^{(i)}, \cdot)$

3. Updating

Likelihood

- For $i = 1, \dots, n$, set $\pi_{t+1,0}^{(i)} \leftarrow g_{t+1}(y_{t+1} - h_{t+1}(\bar{x}_{t+1,0}^{(i)}))$
- For $i = 1, \dots, n$, set $\bar{x}_{t+1,0}^{(i)} \leftarrow \frac{\pi_{t+1,0}^{(i)}}{\sum_{j=1}^n \pi_{t+1,0}^{(j)}}$

4. Resampling

- For $i = 1, \dots, n$, set $x_{t+1,0}^{(i)} \leftarrow \bar{x}_{t+1,0}^{(j)}$ with probability $\pi_{t+1,0}^{(j)}$
- $t \leftarrow t + 1$ and go to step 2

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